SURVEYING & LEVELLING · NOTE SL04

Cross-Staff Survey — Right Angles, Cross Staff and Optical Square

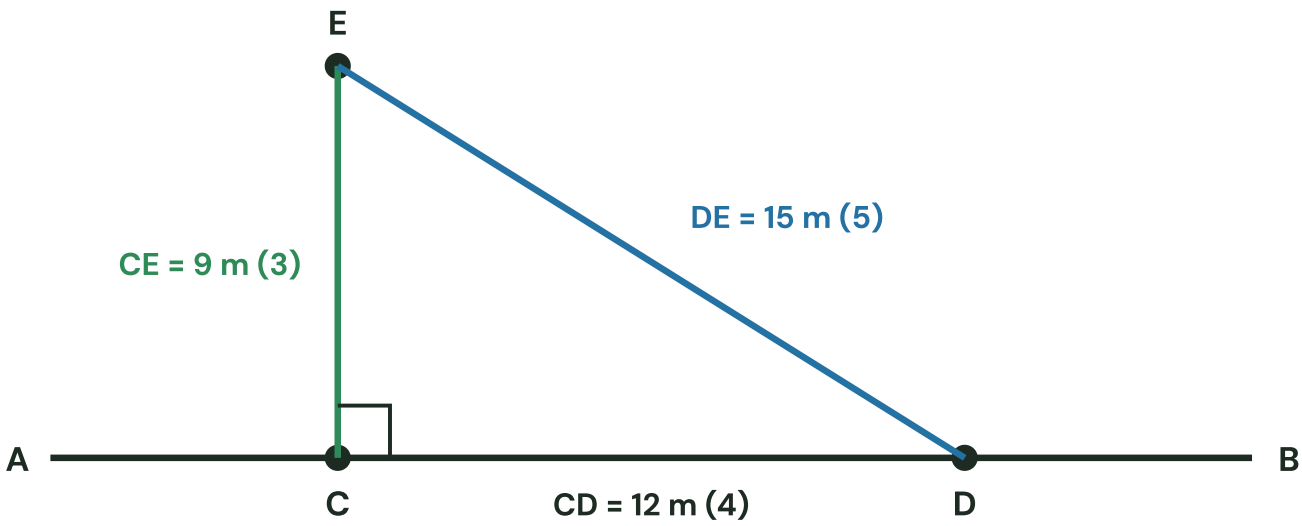
Course topics: SWCE1.4 L5

Setting a right angle with a chain or tape

(a) Three–four–five method

A triangle whose sides are in the ratio 3 : 4 : 5 is right-angled. Common field lengths are 9 m, 12 m and 15 m (or any multiples).

1. Mark point C on the base line AB and measure CD = 12 m along it.
2. With C as centre and radius CE = 9 m, swing an arc.
3. With D as centre and radius DE = 15 m, swing a second arc; the intersection is E.
4. Join CE — it is perpendicular to AB at C. Check all three lengths.



The 3–4–5 (9–12–15 m) right angle

The method is quick and simple and suits ordinary field accuracy.

(b) Chord method

- **From a point C off the line:** with C as centre, swing an arc cutting AB at D and E; find the midpoint F of DE; CF is perpendicular to AB.
- **From a point C on the line:** mark equal distances CD and CE on AB; with D and E as centres swing equal arcs meeting at F; CF is perpendicular.

(c) Semicircle method

1. Take a point D outside AB; with D as centre and DC as radius, draw a semicircle cutting AB at E.
2. Join ED and extend it to meet the semicircle at F.
3. Join CF — it is perpendicular to AB (the angle in a semicircle is 90°).

Measuring an angle without an instrument (sine method)

1. To measure angle CAB, mark equal distances from A on AB and AC: points D and E.
2. Measure DE and compute $\sin(\theta/2) = (DE/2) / AD$.
3. Find $\theta/2$ from sine tables; the angle $\theta = 2 \times (\theta/2)$.

Laying out an angle other than 90°

1. Halve the required angle θ and choose equal lengths AD = AE.
2. Compute $DE = 2\ (AD \times \sin(\theta/2))$ and mark E on AB.
3. One person holds the tape at A, another marks D so that AD and ED have the computed lengths; line AD gives the angle.

Cross staff

- **Uses:** find the foot of a perpendicular; set a right angle at a point on a chain line.
- **Construction:** four metal arms with vertical slits on an iron-shod staff 1.2–1.5 m long. ICAR eCourse: the open cross staff (the common type; others are French and adjustable) sits on a 1.5 m pole of 2.5 cm diameter with an iron shoe.
- **Working:** align one pair of slits with the chain line; the other pair then gives the perpendicular direction.
- Accuracy is moderate.

Optical square

- A small circular box with two mirrors: the **horizon glass** (half-silvered) and the **index glass** (fully silvered).
- It works by double reflection: when the direct and reflected images coincide, the two lines are at 90°.
- It is compact, accurate and fast — more accurate than a cross staff.

Cross-staff survey

- **Principle:** divide the area into right-angled triangles and trapeziums.
- **Instruments:** a chain and a cross staff.
- **Method:** run a base line through the centre of the area and take offsets perpendicular to it on both sides, in the order of chaining.
- **Check:** measure the boundary lengths. Area = sum of the areas of the triangles and trapeziums.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The 3–4–5 method of setting a right angle can use sides of

(a) 6, 8, 12 m
(b) 9, 12, 15 m
(c) 10, 12, 14 m
(d) 5, 12, 15 m

Q2. In an optical square the horizon glass is

(a) fully silvered
(b) half silvered
(c) unsilvered
(d) a prism

Q3. The optical square works on the principle of

(a) refraction
(b) double reflection
(c) diffraction
(d) single reflection

Q4. A cross-staff survey divides the area into

(a) equilateral triangles
(b) right-angled triangles and trapeziums
(c) circles and squares
(d) polygons

Q5. In the sine method, sin(θ/2) equals

(a) DE/AD
(b) (DE/2)/AD
(c) AD/DE
(d) 2DE/AD

Answers

Q1 — (b) 9, 12, 15 m — *9:12:15 = 3:4:5.*
Q2 — (b) half silvered
Q3 — (b) double reflection
Q4 — (b) right-angled triangles and trapeziums
Q5 — (b) (DE/2)/AD